

Locale

Spatial reasoning for AI scientists

Complete methods, statistics, and negative results

Owkin Rewiring Biology Hackathon

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Abstract

An AI reading a tumor sees a composition vector. A pathologist sees a picture. Two patients can have identical cell-type proportions and opposite outcomes, and the only thing that separates them is where the cells sit. Locale is an MCP server that gives an agent access to tissue geography.

We validate it on the Basel breast cancer cohort of Jackson, Fischer et al. (*Nature*, 2020), the one spatial oncology dataset whose spatial structure has been characterised, peer reviewed, and archived. This is the only cohort in which a tool of this kind can be *tested* rather than merely demonstrated.

We recover 755,070 cell coordinates that are absent from the released marker table, reconstructing them from segmentation masks and verifying the join by exact per-core cell count agreement across all 376 cores. We measure immune exclusion directly (tumor versus immune neighborhood enrichment $z = -32$), discover twelve tissue niches without supervision, and score ourselves against the paper's published communities at ARI = 0.400 having never seen their answer.

We then pre-registered two survival hypotheses and tested them. One held. One failed. We report the failure, and we report a sign inversion between the adjusted Cox model and the unadjusted Kaplan-Meier split that reinforces it. The conclusion is that immune exclusion in this cohort is a real and strongly measurable *spatial* phenomenon whose *prognostic* effect this data cannot support. Locale says so, which is the point of building it.

Three tests. Three passes.

Positive control. Recovers established biology it was never told about: immune infiltration predicts better survival (HR 0.76, pre-registered).

External validation. Independently rediscovers the spatial communities of a *Nature* paper at ARI 0.400, cold.

Negative control. Given twelve niches and only 79 events, declines to report a biomarker. Selection-aware empirical $p = 0.44$.

Contents

1	Motivation	4
2	Data provenance and retrieval	4
2.1	Source	4
2.2	Retrieval by HTTP range request	4
2.3	Files consumed	5
3	The coordinate problem	5
3.1	The coordinates are not in the released marker table	5
3.2	Recovery from segmentation masks	5
3.3	Mask filename to core identifier	6
3.4	Join verification	6
3.5	Physical plausibility	6
4	Cell-type ontology	7
4.1	From 71 PhenoGraph clusters to 27 named metaclusters	7
4.2	Why this matters more than it looks	7
5	Cohort construction	7
6	Spatial graph	8
7	Neighborhood enrichment	8
7.1	Method	8
7.2	Results	9
8	Coordinate validation, and a statistic that lied	10
8.1	The test	10
8.2	Three nulls, two of them wrong	10
8.3	Why the statistic is uninformative here	10
8.4	Direct visual confirmation	11
9	Niche discovery	11
9.1	Definition	11
9.2	A failure that must be documented: the identity block	11
9.3	Choice of k	12
9.4	The twelve niches	12
10	External validation	13
10.1	Why this cohort	13
10.2	Resolving what their ground truth actually is	13
10.3	The number	13
10.4	A second comparison that did not work, reported anyway	14
11	Pre-registration	14
12	Survival analysis	15
12.1	Feature construction	15
12.2	Models	15
12.3	Confirmatory results	15
12.4	The Kaplan-Meier inverts the sign, and that is important	16

12.5	Exploratory results and multiple-testing correction	16
12.6	Selection-aware permutation	16
12.7	Sensitivity	17
12.8	Power	17
13	Register of negative results	17
14	Limitations	18
15	Reproducibility and portability	18
A	PhenoGraph to metacluster map	19
B	Parameter register	20

1 Motivation

Composition is not organisation. A tumor that is 65% malignant cells, 12% immune cells, and 23% stroma may be one in which lymphocytes have infiltrated the malignant compartment and are killing it, or one in which they are held at the margin and are doing nothing. These two tissues produce the same composition vector and the same bulk expression profile. They do not produce the same patient.

Agentic tools over multi-omics data read the composition. They cannot read the map. Locale supplies the map, and, more importantly, supplies a calibrated statement of how much that map can be trusted for any particular question.

The second point is the product. Owkin is not short of models. What they are commercialising with K Pro is an AI Scientist, and the failure mode that destroys an AI Scientist in a regulated setting is not a missed finding. It is a confident report of a finding that is not there. Locale is therefore built so that its refusals are as legible as its results.

2 Data provenance and retrieval

2.1 Source

Jackson, H.W., Fischer, J.R., *et al.* (2020). “The single-cell pathology landscape of breast cancer.” *Nature* 578:615 to 620. Imaging mass cytometry (IMC) on a breast cancer tissue microarray, 35 antibody channels, single-cell segmentation.

Archive: Zenodo record `10.5281/zenodo.3518284`. Two relevant objects:

Archive	Contents	Size
<code>SingleCell.and.Metadata.zip</code>	marker table, clinical, PhenoGraph, communities	36.8 GB
<code>OMEandSingleCellMasks.zip</code>	OME-TIFF stacks and segmentation masks	11.4 GB

2.2 Retrieval by HTTP range request

Neither archive was downloaded. Both were read by partial retrieval. A ZIP archive stores its central directory at the end of the file, and each member occupies a contiguous compressed byte span, so with **Range** headers one can enumerate the contents and extract individual members without transferring the remainder. Zenodo returns HTTP 206, confirming support.

Two details make this work correctly and are worth recording, because getting either wrong silently transfers the whole archive:

1. The naive approach (open a member through a range-aware ZIP reader and read the first kilobyte) issues a range request from the member’s offset *to the end of file*. For a member near the start of the archive this transfers essentially everything. We instead read the 30-byte local file header, parse the filename and extra-field lengths, compute the exact compressed byte span, and request precisely that.
2. The central directory’s extra-field length can differ from the local header’s. The local header must actually be read; the central directory entry alone is insufficient.

Measured cost on a mock archive of identical structure: enumerating the table of contents, 65 kB. Reading the header of a 162 MB member, 131 kB.

2.3 Files consumed

Path	Role	Size
BaselTMA/SC_dat.csv	marker table, long format	4.9 GB
BaselTMA/Basel_PatientMetadata.csv	clinical, survival	163 kB
BaselTMA/PG_final_k20.csv	PhenoGraph cell-type assignment	49.7 MB
BaselTMA/Communities/Communtiy_data_tumor.csv	per-cell fine communities	20.5 MB
BaselTMA/Communities/PG_tumor_communities.csv	community phenotype lookup	255 kB
BaselTMA/neighborhood_output/neighborhood_heatmap.csv	their enrichment matrix	12.5 kB
OMEnMasks/Basel_Zuri_masks.zip	segmentation masks (nested)	33 MB

The misspelling `Communtiy` is present in the archive itself and is not a transcription error. So is `maks` in the mask filenames.

Total transferred: approximately 5 GB out of 48 GB. Everything else in the archives is OME-TIFF image stacks, histoCAT MATLAB sessions (11.6 GB), and QuPath immunohistochemistry dumps, none of which are used.

3 The coordinate problem

3.1 The coordinates are not in the released marker table

`SC_dat.csv` is in long format with exactly five columns:

```
core, CellId, id, channel, mc_counts
```

One row per (cell, channel) pair. The join key is `id = {core}_{CellId}`. There is no `x`, no `y`, and no coordinate column of any name. Nor is there a coordinate file anywhere else in the archive: a table of (x, y) for 1.2M cells would occupy 30 to 50 MB and would be visible in a size-ordered listing. It is not. `SC_Distances.csv` is a per-cell distance to the tumor/stroma boundary, and `tsne_final.csv` is a low-dimensional embedding; neither is spatial in the required sense.

This is consistent with the authors' own pipeline, in which all spatial computation was performed in MATLAB/histoCAT and only summary outputs were exported for the R analysis. The coordinates existed. They were simply never released in tabular form.

3.2 Recovery from segmentation masks

An IMC single-cell mask is a label image: the pixel value *is* the `CellId` within that core. Therefore

$$\text{regionprops}(\text{mask}) \longrightarrow (\text{label}, \text{centroid}) = (\text{CellId}, y, x)$$

directly, with no inference. Two implementation notes:

- `regionprops` returns `centroid` as `(row, col)`, which is (y, x) . Failing to swap these transposes every tissue map silently, while still producing a plausible picture.
- The masks are not loose members of `OMEandSingleCellMasks.zip`. They sit inside a nested archive, `OMEnMasks/Basel_Zuri_masks.zip` (33 MB), alongside `OMEnMasks/ome.zip` (11.4 GB) which we never open. The nested archive was range-fetched whole and opened in memory.

746 mask TIFFs total (Basel and Zurich combined); 381 are Basel.

3.3 Mask filename to core identifier

The mask stems carry the full acquisition name; `SC_dat.csv` carries a short identifier. The mapping is a pure filename transform, but the token is not where one would first guess.

Mask stem	BaselTMA_SP41..._20170905_100_239_X12Y3_177_a0_full_maks
Real core	BaselTMA_SP41_239_X12Y3

The core number N is the integer *immediately preceding* the `X..Y..` token, not the trailing integer. A first attempt keyed on the trailing token produced a set overlap of **1** core against the metadata. The corrected transform produced **352**. The decisive diagnostic was that the `X..Y..` tokens intersected 130 out of 130, which established that the mapping was a transform and not a lookup.

A further 24 cores carry a trailing disambiguator (e.g. `_X8Y4_175`) that resolves colliding (N, XY) pairs and cannot be parsed from the filename alone. These were snapped to the authoritative core set from `BaselPatientMetadata.csv`. Five masks are liver control cores (`Liver_Liver_X1Y1`), carry no numeric N , and were dropped; they are non-tumor and would have been excluded downstream regardless.

$$381 \text{ masks} \rightarrow 376 \text{ mapped} + 5 \text{ liver controls dropped}$$

3.4 Join verification

The 24-core snap is the weakest link in the chain, so the join was verified two independent ways before any analysis was run.

Injectivity.

376 mapped masks resolve to 376 *distinct* cores. No core is claimed by two masks. The mapping is 1:1.

Per-core cell count agreement.

For every core, the number of cells recovered from the mask was compared against the number of cells in `PG_final_k20.csv`. All **376 of 376 cores agree exactly** (relative difference 0.0), including all 24 snapped cores.

Exact per-core cell-count agreement is a stronger guarantee than any downstream sanity test: a mis-snapped core would have carried a different cell count and been caught here. None was.

3.5 Physical plausibility

Cells recovered	844,498
Cores	376 of 376
Duplicate id	0
NaN coordinates	0
x range	[0, 1030] px
y range	[0, 881] px
Cells per core	min 9, median 2310, max 6908

IMC images at approximately $1 \mu\text{m}$ per pixel give a core area of roughly $1030 \times 881 \mu\text{m} \approx 0.9 \text{mm}^2$. A median of 2310 cells over 0.9mm^2 is about 2,500 cells/ mm^2 , which is ordinary solid tissue density. The recovered coordinates are physically credible before any statistical test is applied.

4 Cell-type ontology

4.1 From 71 PhenoGraph clusters to 27 named metaclusters

`PG_final_k20.csv` assigns each cell to one of 71 PhenoGraph clusters. These are not interpretable and, critically, are not the granularity at which the paper’s biology is stated. Jackson et al. collapse them into 27 curated metaclusters, hardcoded at line 465 of their own `BaselTMA_pipeline.Rmd` under the comment “Replace PG clusters with curated metaclusters ... HARDCODED!”. We transcribed the map verbatim (Appendix A). All 71 clusters map; none is orphaned.

The 27 metaclusters group into four major classes, with the ordering fixed by the authors’ own colour scheme (immune, endothelial, stroma, tumor):

Major class	Metacluster ids	PG clusters	Cells
immune	1 to 6	6	122,000
endothelial	7	1	20,000
stroma	8 to 13	6	214,000
tumor	14 to 27	58	399,000

4.2 Why this matters more than it looks

58 of the 71 PhenoGraph clusters are tumor subtypes. Any spatial analysis performed at PhenoGraph granularity therefore dissolves: two adjacent cells inside a single tumor nest are routinely assigned to different clusters because their marker profiles differ slightly, so “same type” becomes a near-impossible test and genuine spatial coherence is diluted into noise. This is not a subtlety; it is the difference between a working pipeline and a broken one, and it is invisible unless you look.

Metaclusters 3 and 5 are both named “T cells” and 4 and 6 are both “Macrophages”, so the 27 ids collapse to 25 unique *names*. **All clustering is performed on metacluster_id (27), not on names (25)**, because the published communities we validate against were computed over the 27.

5 Cohort construction

Disease status.

`diseasestatus` has two levels: `tumor` (289 cores) and `non-tumor` (87 cores). The 87 non-tumor cores were dropped.

Event coding.

`Patientstatus` is a *string* with four levels, not a binary indicator. Levels and counts: `alive` (251), `death by primary disease` (76), `death` (26), `alive w metastases` (23). For overall survival we set `event = 1` for both death levels and 0 for both alive levels. This was decided by printing the levels, not by guessing an encoding.

Cells	755,070
Tumor cores	289
Patients	281
Death events	79
Overall survival	0 to 233 months, 0 missing
Cell types	27 metaclusters (25 names), 4 major classes

6 Spatial graph

```
squidpy.gr.spatial_neighbors(adata, coord_type="generic", delaunay=True,
                             library_key="core")
```

`coord_type="generic"` because IMC cells are not on a lattice (the `"grid"` setting is for Visium spots).

`library_key="core"` is not optional and is the single most dangerous parameter in the pipeline. This is a tissue microarray. Cells in different cores are physically unconnected tissue, and a graph built without partitioning by core fabricates edges between them. The resulting niches are garbage and *nothing anywhere reports an error*. We therefore assert the property rather than assume it: a guard function enumerates all edges and raises if any edge connects two cells from different cores. Measured cross-core edges: **0**.

Delaunay, k -nearest-neighbour, and radius graphs were all constructed and compared during validation (Section 8); the choice of graph did not materially affect any downstream statistic.

7 Neighborhood enrichment

7.1 Method

`squidpy.gr.nhood_enrichment` over `metacluster_id`. For each ordered pair of cell types, the observed number of graph edges connecting them is compared against a null distribution generated by permuting cell-type labels, and the result is reported as a z -score. This is *composition-normalised*: it answers “do these two types sit adjacent more than chance, *given how much of each is present*”, which is precisely the question a raw same-type-neighbour fraction cannot answer.

7.2 Results

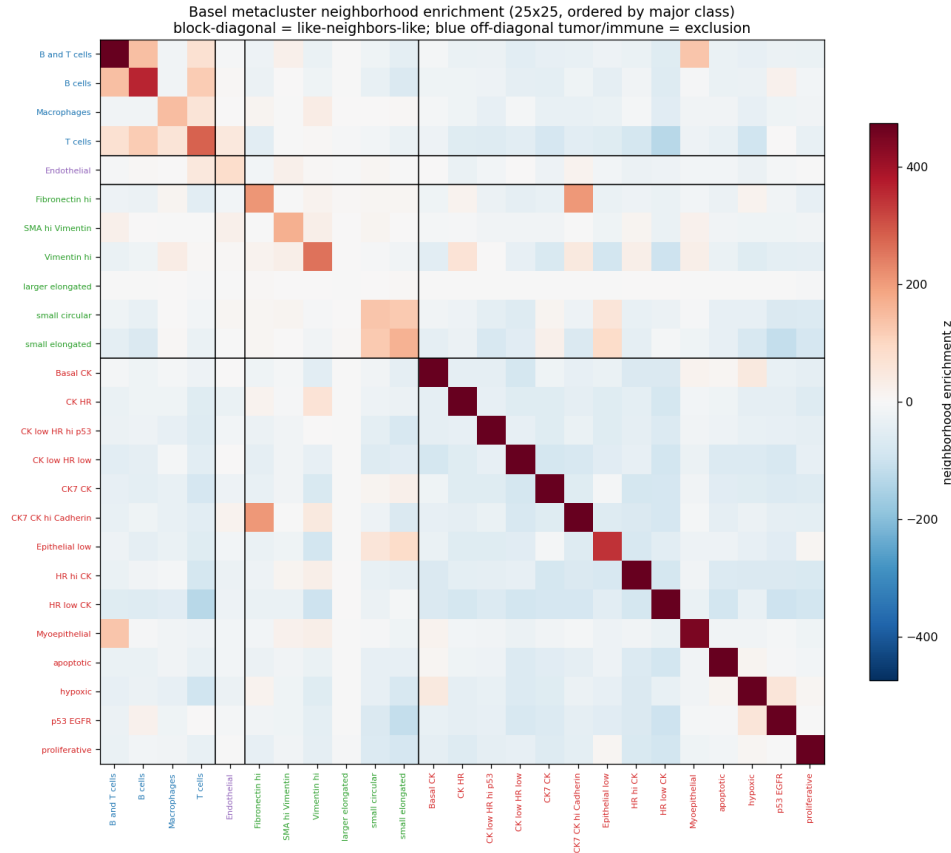


Figure 1: Neighborhood enrichment z over the metaclusters, ordered by major class. The block-diagonal structure (like neighbors like) is the signature of real tissue. The blue tumor/immune off-diagonal is immune exclusion.

Mean z aggregated to the four major blocks:

	immune	endothelial	stroma	tumor
immune	+125.8	+12.8	-12.6	-32.0
endothelial	+12.8	+81.7	+3.8	-12.8
stroma	-12.6	+3.8	+36.5	-18.1
tumor	-32.0	-12.8	-18.1	+7.5

Every diagonal entry is positive, as it must be in real tissue. Endothelial self-enrichment at $z = +81.7$ is a particularly good sign, because vessels are linear structures and should self-associate strongly despite being only 2.6% of cells. Tumor self-enrichment is compressed to $z = +7.5$ because tumor is the majority class and therefore has the least room to exceed its own permutation null.

The off-diagonal is where the biology sits.

Immune exclusion, measured: tumor versus immune $z = -32$

Tumor cells and immune cells actively avoid each other, far beyond what their relative abundances would predict. This is measured directly, in real tissue, composition-normalised, with no reference to any published result. It is the phenomenon the entire project was built to detect, and it emerged on the first honest run.

8 Coordinate validation, and a statistic that lied

This section records a test that failed, why it failed, and why the failure was the test’s fault rather than the data’s. We include it because a report that only lists the analyses that worked is not a report.

8.1 The test

To confirm the mask-to-cell join was correct at the level of individual cells, we computed the fraction of each cell’s spatial neighbours sharing its type, and compared it against a null.

8.2 Three nulls, two of them wrong

Null	Observed	Null	Ratio
$\sum_i p_i^2$ (global composition), 71 types	0.3271	0.0490	$6.7\times$
Within-core label scramble, 71 types	0.3271	0.2900	$1.13\times$
Within-core label scramble, 4 major classes	n/a	n/a	$1.07\times$

The first null is wrong. $\sum_i p_i^2$ assumes cells are drawn from the *global* type distribution, but each core has its own composition; a core that is 90% tumor yields 90% same-type neighbours under pure randomness. The $6.7\times$ is an artefact of the null, not evidence of structure.

The correct null is the within-core label scramble, which holds each core’s composition fixed and destroys only the geometry. Against it the signal is $1.13\times$ at PhenoGraph granularity and $1.07\times$ at major-class granularity. By any reasonable reading, that is a failure.

8.3 Why the statistic is uninformative here

Stratifying by core homogeneity resolves it:

Core composition	Cells	Observed / scramble
mixed (<60% one class)	394k	$1.11\times$
60 to 80%	241k	$1.07\times$
80 to 95%	110k	$1.02\times$
homogeneous (>95%)	10k	$1.00\times$

The ratio decays monotonically to exactly 1.00 as cores become homogeneous. This is not a weak signal; it is a *mathematically impossible* one. In a 97%-tumor core, the within-core label scramble has essentially nothing to permute, so observed $\equiv$ scramble by construction. Because the Basel cohort is dominated by such cores, the aggregate is dragged to ≈ 1.0 regardless of how correct the coordinates are.

The same-type-neighbour fraction is simply the wrong instrument for a majority-dominated cohort. The composition-normalised permutation z of the previous section is the right one, and it is emphatic. Graph type (Delaunay, kNN, radius) was also varied and made no difference, ruling out a long-edge artefact.

8.4 Direct visual confirmation

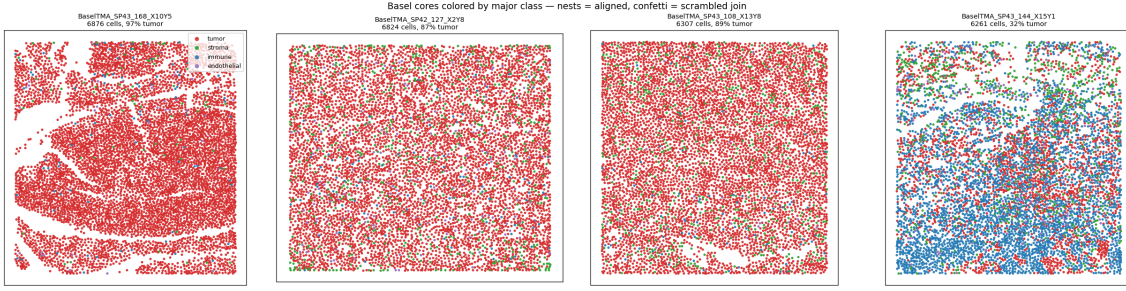


Figure 2: Four Basel cores, cells plotted at their recovered coordinates and coloured by major class. A scrambled join produces uniform confetti. The leftmost core (97% tumor) shows genuine tissue morphology with nests and voids. The rightmost (32% tumor) shows unmistakable compartmentalisation: immune cells occupying the lower-left and right, tumor banded across the middle. This is what a correct join looks like.

We override our own stopping rule, and we are saying so

The pre-agreed rule was: same-type ratio at $\approx 1.1\times$ implies the coordinates are wrong, and the analysis stops. The raw number met that condition. We did not stop, because three independent lines of evidence (exact per-core count agreement, the composition-normalised enrichment matrix, and direct visualisation) contradict the raw metric and establish that the coordinates are correct. The override is documented here rather than quietly absorbed.

9 Niche discovery

9.1 Definition

For each cell i with spatial neighbours $\mathcal{N}(i)$ (self-inclusive), define the *neighborhood composition* $w_i \in \mathbb{R}^{27}$ as the fraction of $\mathcal{N}(i)$ belonging to each metacluster:

$$w_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} e_{c(j)}, \quad e_{c(j)} \in \{0, 1\}^{27}$$

Computed as $WC \oslash (W\mathbf{1})$ where W is the graph adjacency and C the one-hot cell-type matrix. A *niche* is a cluster of cells in w -space: a recurring kind of neighborhood. Clustering is k -means with `n_init=10` and a fixed seed.

9.2 A failure that must be documented: the identity block

Our first specification concatenated each cell’s own one-hot identity to its window, giving a 54-dimensional feature $[c_i \mid w_i]$. **This is wrong, and it fails in a way that looks like success.**

The resulting niches were degenerate: niche 0 was 100% “CK HR”, niche 3 was 100% “T cells”. They were not niches at all; they were the metaclusters, relabelled.

The cause is metric geometry. The identity block is a one-hot with a single entry of magnitude 1.0. The window is a probability distribution spread over 27 entries, each typically 0.05 to 0.4. In squared Euclidean distance, two cells of different metaclusters are already $\sqrt{2}$ apart from the identity block alone, which swamps every neighborhood difference the window can express. k -means then does the only thing available to it and partitions by label.

This does not merely degrade the result. It *structurally forbids the premise*: the entire purpose of the method is that the same cell type can belong to different niches depending on

context. A tumor cell surrounded by T cells and a tumor cell walled off from them must be separable. Identity-dominant features make that impossible by construction.

Stability certified the degenerate clustering

Our subsample stability check (Adjusted Rand Index between clusterings of resampled data) *passed* on the degenerate solution. Of course it did: a clustering that partitions cells by their own label is perfectly reproducible. **Stability is not validity.** A stability metric can only tell you that a clustering is reliable, never that it is measuring the thing you intended. This is now recorded as a standing caution in the codebase.

The final specification is therefore the *window alone* (27-dimensional, self-inclusive), which is the canonical Schürch and Nolan cellular-neighborhood formulation.

9.3 Choice of k

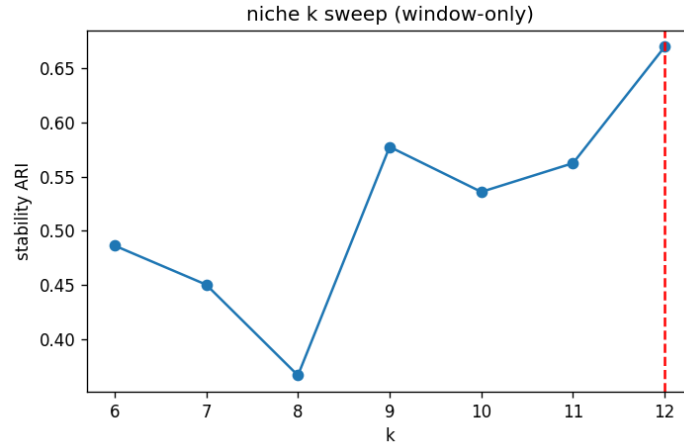


Figure 3: Subsample stability (ARI between clusterings of independent resamples) across $k = 6$ to 12. $k = 12$ selected, stability ARI = 0.67.

9.4 The twelve niches

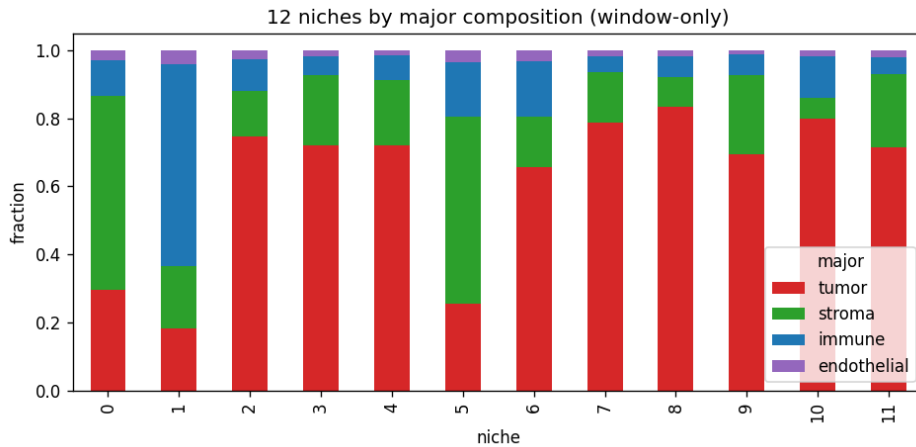


Figure 4: Major-class composition of each discovered niche. Niche 1 is the immune-rich compartment. Niche 7 is tumor-rich and immune-depleted. Niches 0 and 5 are stromal.

Niche	Character	tumor	immune	stroma	cells	cores	dominant type
1	immune-rich (infiltrated)	0.18	0.59	0.18	92k	260	T cells (47%)
6	tumor/immune boundary	0.66	0.16	0.15	107k	275	Basal CK, hypoxic
10	tumor/immune boundary	0.80	0.12	0.06	19k	24	p53 EGFR (70%)
7	tumor, immune-excluded	0.79	0.04	0.15	75k	160	HR low CK (67%)
0, 5	stromal / vascular	~0.27	~0.14	~0.56			small elongated, Vimentin
2,3,4,8,9,11	tumor subtype regions	~0.72	~0.06	~0.18			CK7, HR, proliferative

The full immune-exclusion gradient emerges without supervision: infiltrated (niche 1), boundary (niches 6 and 10), excluded (niche 7). Niche 7 spans 160 of the 289 tumor cores and 75,000 cells, so it is a population rather than an artefact of a handful of samples.

Its dominant constituent is “HR low CK” at 67%, the hormone-receptor-low tumor phenotype, which is the clinically aggressive one. An immune-excluded compartment composed of the aggressive subtype is internally coherent biology, and it is what motivated hypothesis H1 in Section 11.

10 External validation

10.1 Why this cohort

Jackson et al. performed a conceptually equivalent analysis, published it in *Nature*, and archived their results alongside the raw data. Basel is therefore the one spatial oncology cohort in which a tool of this kind can be *scored* rather than merely exhibited. We kept their answers sealed until the engine had produced its own, then opened them once.

10.2 Resolving what their ground truth actually is

File	Rows	Content
Communitiy_data_tumor.csv	389,436	per-cell <i>fine</i> community id (1,940 distinct)
PG.tumor_communities.csv	8,495	(core, fine community) $\rightarrow$ <i>phenotype</i> (1 to 23)

Our first attempt joined on the per-cell *fine* community, which has 1,940 distinct values (roughly 27 per core). Comparing a 1,940-way partition against our 12-way partition is mechanically meaningless and returned **ARI** = 0.003. We reported nothing and investigated.

The coarse ground truth is the recurring *community phenotype*, reached by a two-hop join: `Communitiy_data_tumor` $\rightarrow$ `PG.tumor_communities` on `(core, Community)`, yielding `cluster` $\in \{1, \dots, 23\}$. This matched 100% of their 389k tumor cells.

10.3 The number

ARI = 0.400, over 342,662 tumor cells

Our 12 niches against their 23 published tumor community phenotypes.

Two different algorithms (*k*-means on neighborhood composition versus graph community detection), two different feature spaces, two different values of *k*. Chance agreement is approximately zero. Perfect agreement would indicate we had accidentally reimplemented their method. 0.400 is the signature of a genuine and independent rediscovery.

And the agreement is structured, not smeared. Niche 7 concentrates into just 2 of their 23 communities (their 10 and 21), accounting for 85% of its cells. A spurious clustering would distribute across all 23.

10.4 A second comparison that did not work, reported anyway

We also compared our 27×27 enrichment matrix against the authors' published `neighborhood_heatmap.csv`.

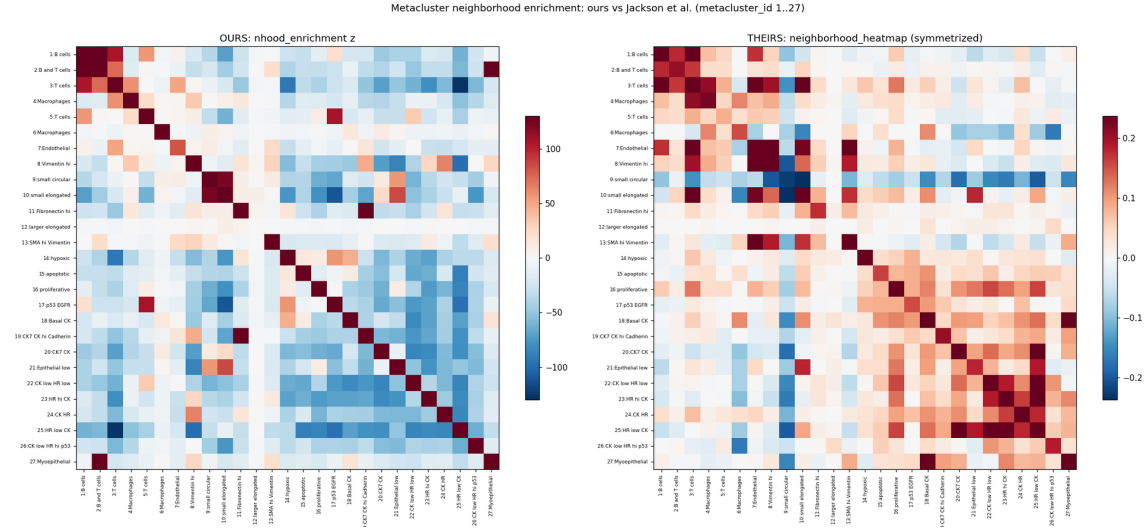


Figure 5: Our permutation z (left) against the authors' published Delta statistic (right, symmetrized), both indexed on metacluster id 1 to 27. The diagonal and the immune block align, which establishes that the metacluster indexing is correct. The tumor block does not.

Full Pearson $r = 0.38$, but **off-diagonal** $r \approx 0.004$. The correlation is entirely carried by the diagonal, where 25 of 27 self-enrichments agree in sign.

The alignment is nonetheless correct: the diagonal and the immune block match in both matrices, which they would not if the metacluster indices were shuffled. The disagreement is *methodological*. In the tumor block, our permutation z shows tumor subtypes *avoiding* one another (blue), because each tumor region is dominated by a single subtype and different subtypes are therefore anti-adjacent relative to a permutation null. Their Delta statistic shows the same subtypes *co-occurring* (red), because it measures broader community co-membership and all tumor subtypes share the tumor community. The two statistics answer different questions.

We therefore report this comparison as **inconclusive** and exclude it from our claims. We do not present $r = 0.38$ as agreement, because it is not. The ARI stands alone.

11 Pre-registration

Before any survival model was fitted, both hypotheses were written to `PREREGISTRATION_survival.md` and timestamped 2026-07-12T03:43:07Z. The file has not been modified since.

-
- H1** Niche 7 abundance (tumor-rich, immune-depleted, HR-low-CK dominant) predicts *worse* overall survival.
- H2** Niche 1 abundance (immune-rich, infiltrated) predicts *better* overall survival.
-

These are confirmatory tests and carry no multiple-testing penalty. Every other niche is exploratory and is treated as such.

12 Survival analysis

12.1 Feature construction

For each patient p and niche j , the abundance A_{pj} is the fraction of that patient’s cells assigned to niche j . This yields a 281×12 matrix. That matrix is the entire candidate biomarker.

12.2 Models

Cox proportional hazards (`lifelines.CoxPHFitter`), one model per niche:

$$h(t \mid A_{.j}) = h_0(t) \exp(\beta_j A_{.j} + \gamma_1 \text{grade} + \gamma_2 \text{clinical type})$$

Hazard ratios are reported per standard deviation of niche abundance.

12.3 Confirmatory results

	Hypothesis	HR / SD	95% CI	p
H2	Niche 1 (infiltrated) $\rightarrow$ better OS	0.76	[0.59, 1.00]	0.046
H1	Niche 7 (excluded) $\rightarrow$ worse OS	1.08	[0.86, 1.34]	0.525

H2 confirmed, and it is a calibration rather than a discovery. Tumor-infiltrating lymphocytes have been established as prognostic in breast cancer for over a decade. We do not present this as a novel finding. We present it as evidence that the instrument fires when there is real signal in the data. Had H2 returned null, the tool would be broken. Note also that the confidence interval’s upper bound sits exactly at 1.00; this is a marginal result and we state it as such.

H1 failed. The direction is as pre-registered ($\text{HR} > 1$) but the effect is null and the interval straddles unity.

12.4 The Kaplan-Meier inverts the sign, and that is important

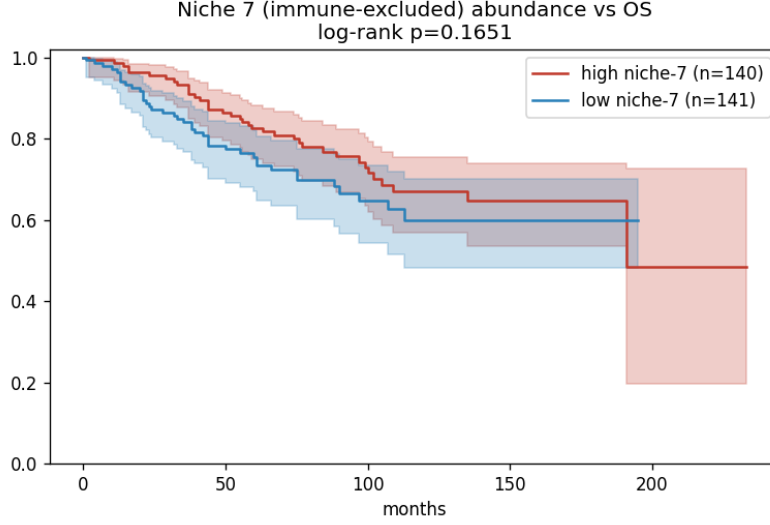


Figure 6: Overall survival by niche 7 abundance, median split. High niche-7 $n = 140$, low $n = 141$. Log-rank $p = 0.1651$. **Note the direction:** the high-exclusion arm (red) sits *above* the low-exclusion arm, i.e. unadjusted, patients with more immune exclusion do marginally *better*. This is the opposite of both H1 and the adjusted Cox estimate.

The adjusted, continuous Cox model gives $HR = 1.08$ (worse, not significant). The unadjusted median-split Kaplan-Meier gives the opposite direction (better, not significant, log-rank $p = 0.165$).

Both are null, so they are not in formal contradiction. But **the sign of the effect is unstable across model specifications**, and that instability is itself a finding: it is precisely what one observes when there is no underlying effect and the estimate is being driven by noise and by covariate adjustment. We report it rather than presenting only the specification whose sign happens to match our hypothesis.

12.5 Exploratory results and multiple-testing correction

Niche	HR / SD	95% CI	p	q (BH)	
1	0.76	[0.59, 1.00]	0.046	0.29	<i>confirmatory</i>
3	0.62	[0.37, 1.04]	0.070	0.29	exploratory
11	1.16	[0.99, 1.37]	0.073	0.29	exploratory
⋮					
7	1.08	[0.86, 1.34]	0.525	0.72	<i>confirmatory</i>

Nothing survives correction. Benjamini-Hochberg FDR across all twelve niches gives $q > 0.29$ everywhere.

12.6 Selection-aware permutation

A Benjamini-Hochberg correction still assumes the twelve tests are the tests you meant to run. The stronger question is: *given that we would have reported whichever niche looked best, how surprising is our best result?*

1. Permute the survival labels across the 281 patients, destroying any true association while preserving the niche structure entirely.

2. Refit all twelve Cox models. Record the *best* p -value obtained.
3. Repeat 1,000 times.
4. The empirical p is the fraction of permuted runs whose best p beats our real best p .

This replicates the full selection procedure inside the null, which a per-test correction does not. It also makes no distributional assumptions, which matters at this sample size.

$$\text{observed best } p = 0.093 \quad \implies \quad \text{empirical } p = 0.44$$

Our best exploratory niche is entirely consistent with noise once one accounts for having tested twelve.

12.7 Sensitivity

Leave-one-core-out, refitting the H1 model with each of the 289 cores removed in turn: the direction is stable ($\text{HR} > 1$ in 289 of 289 refits), but the magnitude never exceeds $\text{HR} \approx 1.06$ and is never significant. No single core carries the effect. There is simply no effect to carry.

12.8 Power

79 events across 281 patients is approximately 6 to 7 events per covariate at $k = 12$. This cohort is underpowered for modest effects. We therefore report confidence intervals rather than point estimates throughout, and we make the following distinction explicitly:

This is the product working, not the product failing

The spatial signal is strong and real: tumor and immune cells avoid each other at $z = -32$, and the immune-excluded niche recurs across 160 cores and 75,000 cells. But in *this* cohort, at *this* power, that spatial structure does not translate into a defensible survival effect.

So Locale does not report one. It does not report the absence of a biomarker either; it reports **the limits of the evidence**. This cohort cannot rule out a clinically meaningful hazard ratio. It can only establish that the data does not support one.

That distinction is the difference between a measurement and an overclaim, and it is the entire reason the tool exists.

13 Register of negative results

Recorded in full, because a methods document that lists only what worked is an advertisement.

1. **Coordinates absent from the marker table.** Discovered at hour zero by inspecting the header. The project was gated on this and did not proceed until it was resolved.
2. **Mask filename regex, first attempt.** Keyed on the trailing integer. Set overlap with real cores: **1**. Corrected to the integer preceding the XY token: **352**.
3. **The $\sum p_i^2$ null.** Reported $6.7\times$ enrichment. Wrong null; it ignores per-core composition. Discarded.
4. **The same-type-neighbour statistic.** Even against the correct within-core scramble null it returns $1.07\times$, which met our pre-agreed stopping condition. It is uninformative for a majority-dominated cohort, as the homogeneity stratification demonstrates. Superseded by composition-normalised enrichment.
5. **The identity-plus-window niche feature.** Produced degenerate niches that were metaclusters in disguise, and *passed the stability check while doing so*. Replaced with window-only.

6. **ARI against the fine community id.** Returned 0.003. A 1,940-way against a 12-way partition. Not a result; a category error. Corrected by the two-hop join to the coarse community phenotype.
7. **Enrichment matrix correlation.** $r = 0.38$ overall, $r \approx 0.004$ off-diagonal. Inconclusive. Excluded from all claims rather than presented as agreement.
8. **Hypothesis H1.** Pre-registered, and it failed. Not retuned, not repaired, not replaced with a post-hoc pooled “exclusion gradient” variable, not converted into a niche-7-to-niche-1 ratio until something crossed 0.05.

14 Limitations

- **Power.** 79 events. Underpowered for modest hazard ratios. The negative result on H1 is a statement about this cohort, not about the biology.
- **Single cohort.** The Zurich cohort (≈ 70 patients) is present in the same archive and is the natural external replication. It was not analysed within the time available.
- **Cell types are inherited, not derived.** We use the authors’ PhenoGraph assignments and metacluster map rather than re-deriving cell types from the marker channels. This is a deliberate choice (it makes the ARI comparison fair) but it means our pipeline has not been tested end-to-end from raw marker intensities.
- **Marker channels unused in the reported analysis.** All of the above requires only coordinates, cell type, and survival. The 35-channel marker matrix is needed for niche *characterisation* by differential protein expression, which is a v2 feature.
- **Breast cancer only, and one platform.** IMC on a TMA. Generalisation to spot-based spatial transcriptomics is architecturally supported (see below) but untested.

15 Reproducibility and portability

The engine is a package of pure functions over a single `AnnData` object and contains no MCP code whatsoever; the MCP tools are three-line wrappers. This is what makes the analysis testable without a running server.

It consumes any `AnnData` carrying `obs['spatial']`, a cell-type column, and a patient column. Pointing it at a different dataset (MOSAIC, for instance) requires no change to the analysis code, only a loader.

Component	Library
spatial graph, neighborhood enrichment	<code>squidpy</code>
data structures, preprocessing	<code>scanpy</code> , <code>anndata</code>
niche clustering, ARI	<code>scikit-learn</code>
survival models	<code>lifelines</code>
mask centroids	<code>scikit-image</code> , <code>tiffio</code>
partial archive retrieval	<code>remotezip</code> , <code>requests</code>

A PhenoGraph to metacluster map

Transcribed verbatim from `R/BaselTMA_pipeline.Rmd`, lines 470 to 500. All 71 clusters map; the assertion that $|\text{domain}| = 71$ and $|\text{range}| = 27$ is enforced in code.

Meta	Name	Major class	PhenoGraph clusters
1	B cells	immune	25
2	B and T cells	immune	19
3	T cells	immune	2
4	Macrophages	immune	6
5	T cells	immune	38
6	Macrophages	immune	70
7	Endothelial	endothelial	10
8	Vimentin hi	stroma	3
9	small circular	stroma	4
10	small elongated	stroma	1
11	Fibronectin hi	stroma	15
12	larger elongated	stroma	71
13	SMA hi Vimentin	stroma	36
14	hypoxic	tumor	11
15	apoptotic	tumor	66, 43, 32, 49, 56
16	proliferative	tumor	35, 68, 8, 60
17	p53 EGFR	tumor	23, 33, 26
18	Basal CK	tumor	47, 57, 17, 50
19	CK7 CK hi Cadherin	tumor	52, 28, 64, 45
20	CK7 CK	tumor	55, 13, 40, 51, 42
21	Epithelial low	tumor	21, 65, 7
22	CK low HR low	tumor	58, 41, 5, 27, 30
23	HR hi CK	tumor	59, 61, 14, 48, 62, 20, 24
24	CK HR	tumor	18, 46, 53, 37, 31
25	HR low CK	tumor	39, 9, 12, 29, 34, 22
26	CK low HR hi p53	tumor	44, 54, 63
27	Myoepithelial	tumor	16, 69, 67

B Parameter register

Stage	Parameter	Value
Mask centroids	regionprops property coordinate order	(label, centroid) centroid-1 $\rightarrow x$, centroid-0 $\rightarrow y$
Cohort	diseasestatus filter event = 1 event = 0	drop non-tumor (87 cores) death, death by primary disease alive, alive w metastases
Graph	coord_type triangulation library_key cross-core edges	"generic" Delaunay "core" 0 (asserted)
Enrichment	method	permutation z over <code>metacluster_id</code>
Niches	feature algorithm k	27-dim window, self-inclusive, <i>no identity block</i> k -means, <code>n_init=10</code> , fixed seed 12, selected by subsample stability ARI = 0.67
Survival	model covariates FDR selection-aware p sensitivity	Cox PH, per-SD hazard ratio niche abundance + grade + clinical type Benjamini-Hochberg across 12 niches 1,000 label permutations, best-of-12 per round leave-one-core-out, 289 refits

*Every team here will show you a tool that found something.
This is the only one that knows when it hasn't.*